

E7 Introduction to Computer Programming for Scientists and Engineers

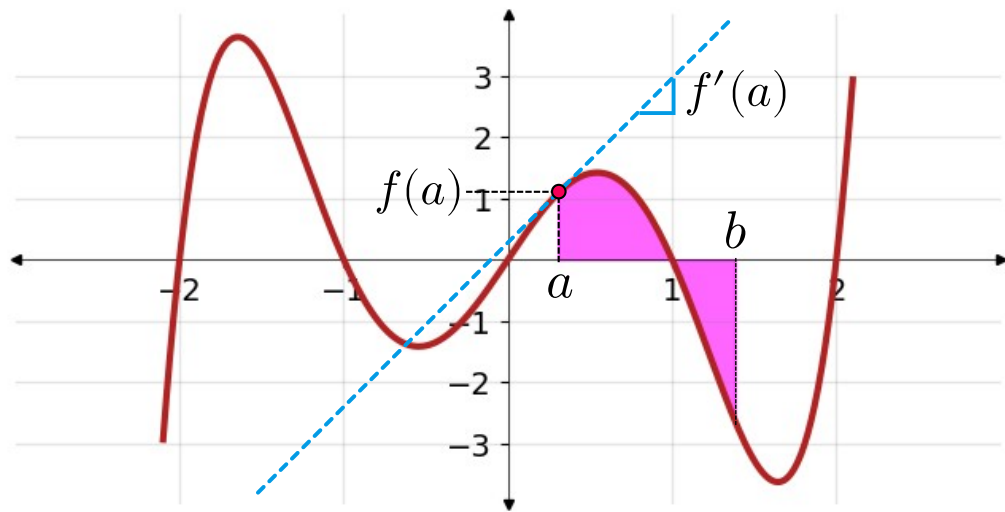
Lecture week 11-A

Numerical differentiation

Recall: Single variable calculus

- Derivative of $f(x)$:

$\frac{df}{dx}(a)$ or $f'(a)$ is the
slope of $f(x)$ at $x = a$

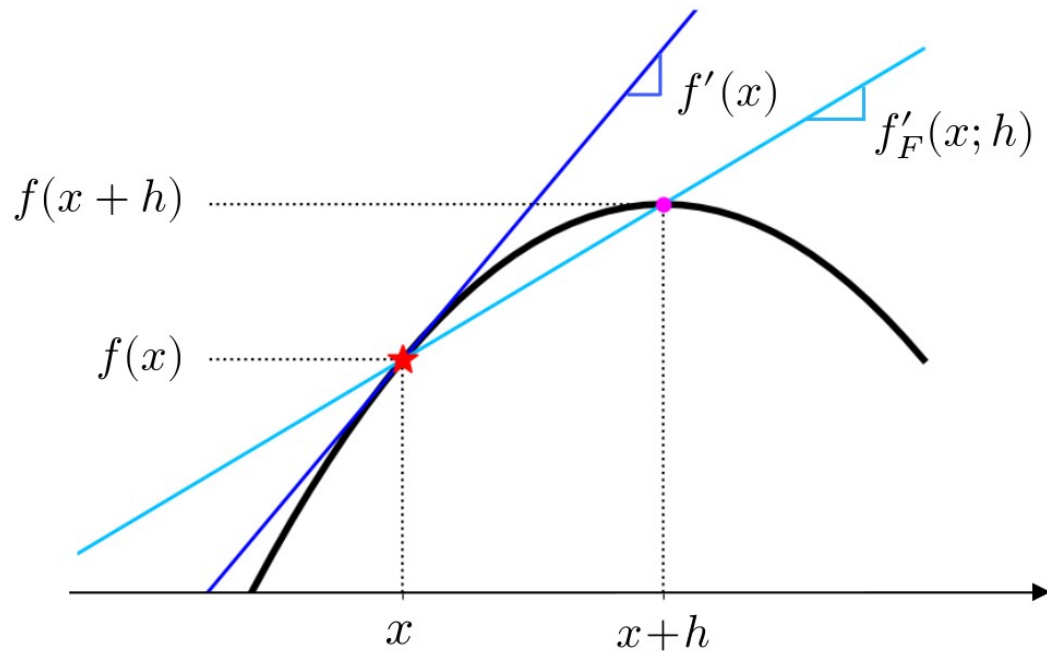


- Integral of $f(x)$ on the interval $[a, b]$:

$\int_a^b f(x) dx$ is the signed “area under the curve”
of $f(x)$ between a and b .

Approximating the derivative with finite differences

$$f'_F(x; h) = \frac{f(x + h) - f(x)}{h} = f'(x) + E_F(x; h)$$



Taylor's approximation of a function at a point

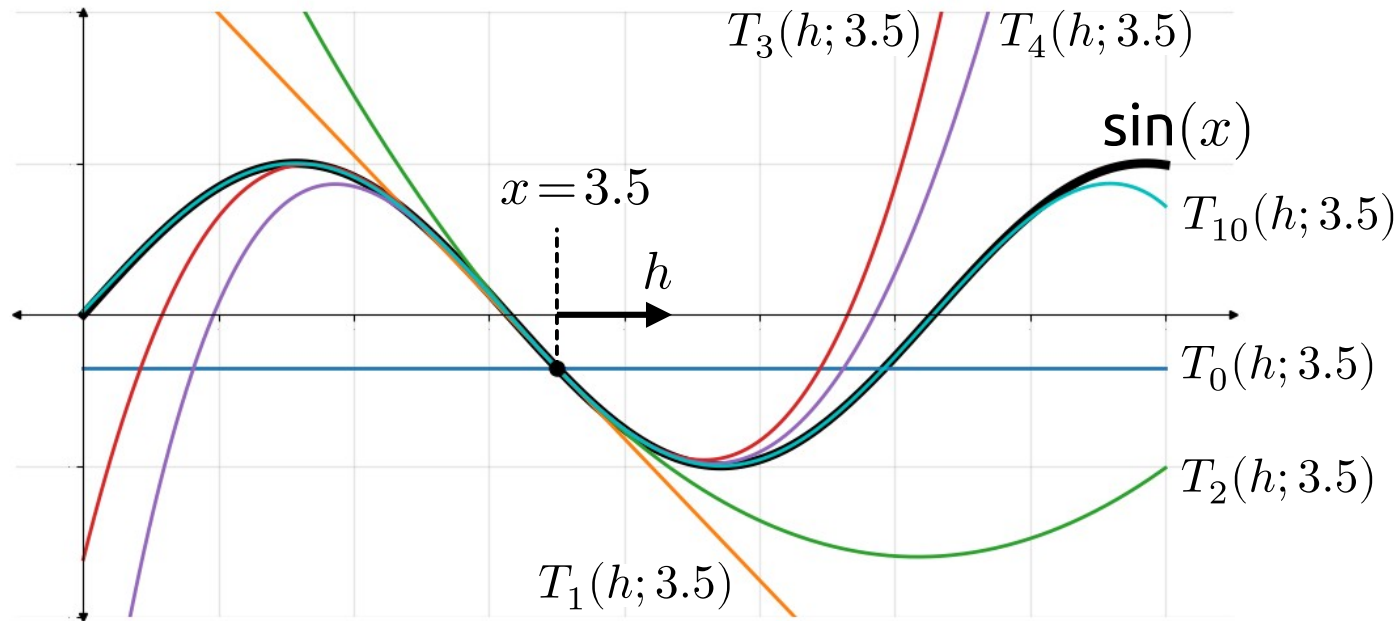
- $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function with at least k derivatives.
- The k 'th order Taylor approximation of f about $x \in \mathbb{R}$ is a k 'th order polynomial in h , denoted $T_k(h; x)$.

$$\begin{aligned} f(x+h) &\approx T_k(h; x) = \sum_{i=0}^k \frac{f^{(i)}(x)}{i!} h^i \\ &= f(x) + f'(x)h + \frac{f''(x)}{2!}h^2 + \dots + \frac{f^{(k)}(x)}{k!}h^k \end{aligned}$$

- Note: $T_k(h; x)$ is a polynomial in h , with x held fixed.

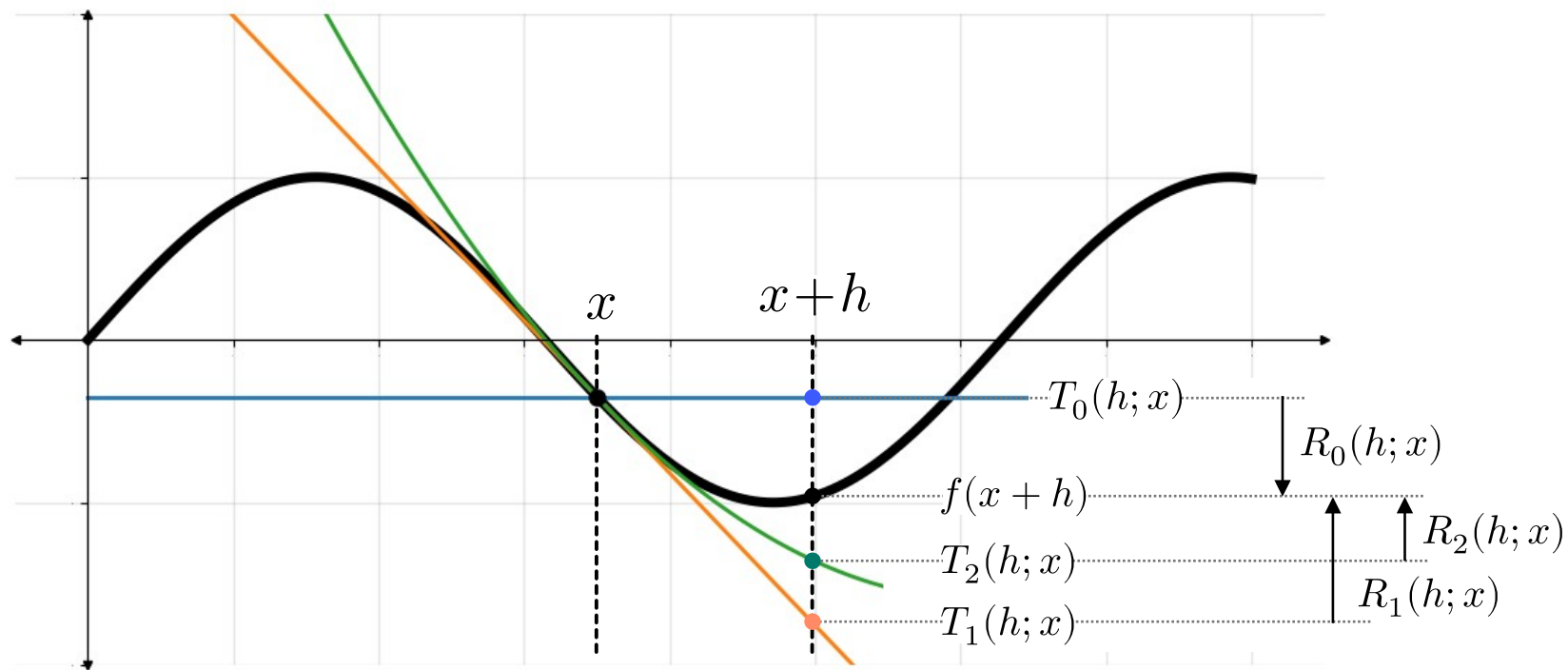
Example: Use a Taylor polynomial to approximate $\sin(x)$ near $x = 3.5$.

$$\sin(3.5+h) \approx \sin(3.5) + \cos(3.5)h - \frac{\sin(3.5)}{2!}h^2 - \frac{\cos(3.5)}{3!}h^3 + \dots$$



- $R_k(h; x)$ is the *approximation error* (a.k.a. the *residual*).

$$R_k(h; x) = f(x+h) - T_k(h; x)$$



Taylor's theorem

For any h and any k , there exists a ξ between x and $x+h$ such that

$$R_k(h; x) = \frac{f^{(k+1)}(\xi)}{(k+1)!} h^{k+1}$$

In particular,

$$|R_k(h; x)| \leq \frac{\max_{\xi \in [x, x+h]} |f^{(k+1)}(\xi)|}{(k+1)!} |h|^{k+1}$$

Bound the error of the forward differentiation scheme

- Apply Taylor's theorem to $f(x)$ with $k=1$, and $h>0$.

$$f(x+h) = f(x) + f'(x)h + \frac{f''(\xi)}{2}h^2 \quad \text{for some } \xi \in [x, x+h]$$

- Rearrange and take absolute value.

$$\left| \frac{f(x+h) - f(x)}{h} - f'(x) \right| = \frac{|f''(\xi)|}{2} h \quad \text{for some } \xi \in [x, x+h]$$

- The left hand side is the approximation error of $f'_F(x; h)$.

$$|E_F(x; h)| = \frac{|f''(\xi)|}{2} h \quad \text{for some } \xi \in [x, x+h]$$

- Since we don't know ξ , let' be conservative.

$$|E_F(x; h)| \leq \frac{\max_{\xi \in [x, x+h]} |f''(\xi)|}{2} h \leq \frac{M_2}{2} h \quad \dots O(h)$$

$$\text{where } M_2 = \max_{\xi \in \mathbb{R}} |f''(\xi)|$$

Other differencing schemes

- Backward difference: $f'_B(x; h) = \frac{f(x) - f(x - h)}{h}$

$$|E_B(h)| \leq \frac{M_2}{2} h \quad \dots O(h)$$

- Central difference: $f'_C(x; h) = \frac{f(x + h) - f(x - h)}{2h}$

$$|E_C(h)| \leq \frac{M_3}{6} h^2 \quad \dots O(h^2)$$

where $M_3 = \max_{\xi \in \mathbb{R}} |f'''(\xi)|$

Approximations for higher order derivatives

- Second derivative: $f_C''(x; h) = \frac{f(x + h) - 2f(x) + f(x - h)}{h^2}$

$$E_C(h) \leq \frac{M_4}{12} h^2$$

- In general, any $f^{(n)}(x)$ can be approximated using points around x .
- The more points, the better the estimate.
- See <https://web.media.mit.edu/~crtaylor/calculator.html>